



Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.

UNIT 2 • STANDARD 6.DS.5

Represent and Describe Data in a Histogram

Lesson 2-2 • Teacher Copy

Name: _____

Date: _____

Class: _____

My Notes

Level 3 Enrichment



TODAY'S OBJECTIVES

Content Objective: I can make and read a histogram to display data in intervals.

Language Objective: I can explain my histogram using the words histogram, frequency, interval, and distribution.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Display Data with Histograms

Histogram

Frequency

Interval

Distribution

Variability



Tap any word to see what it means and a picture.

Level 3: try the blanks from memory first, then check the bank.

1

A histogram groups numerical data into equal and shows each one's .

2

A bar graph that shows how often data falls into equal intervals is a .

3

How many data values fall in an interval is the .

4

An equal-sized range of values used to group data is an .

5 The overall pattern of how data is spread out is the .

6 How much the data values differ is the .

Write About the Math

Guided ESOL writing

Use the support level you need. Every level answers the same math question.

1. Understand the Question

EXPLAIN

Does the histogram support the coach's claim? What does the shape of the data tell you?

Your job: Answer the question. Use math evidence. Explain how the evidence proves your answer.

Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.

2. Plan Your Math Words

Check at least two words you will use.

- ☐ **Display Data with Histograms** — Group numerical data into intervals and show each interval's frequency with a bar.
Representar datos con histogramas — Agrupar datos numéricos en intervalos y mostrar la frecuencia de cada intervalo con una barra.

- ☐ **Histogram** — A bar graph that groups data into equal ranges. The bars touch.
Histograma — Una gráfica de barras que agrupa datos en rangos iguales. Las barras se tocan.

- ☐ **Frequency** — How many times a value shows up.
Frecuencia — Cuántas veces aparece un valor.

- ☐ **Interval** — A range of numbers used to group data.
Intervalo — Un rango de números para agrupar datos.

- ☐ **Distribution** — How the data is spread out.
Distribución — Cómo están repartidos los datos.

Say it first: Say your idea to a partner or quietly to yourself before you write.

Di tu idea a un compañero o en voz baja antes de escribir.

The histogram peaks at the ____ interval with ____ players. The shape is approximately _____. This _____ the coach's claim because _____.

Di tu respuesta primero: Mi respuesta es _____ porque _____.

3. Build Your Explanation

Model the parts: This model shows the parts of an explanation. It does not answer your writing question.

Claim: A histogram groups data into equal intervals so large data sets become readable.

Evidence: Students explain grouping into intervals makes 30 values readable and that each bar's height is the frequency for that interval.

Reasoning: This evidence connects the definition of histogram to the mathematical claim.

Start

Most language support

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- The histogram peaks at the ____ interval with ____ players. The shape is approximately _____. This ____ the coach's claim because _____.
- My answer is _____.

Mi respuesta es _____.

Build

Some language support

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- My claim is _____.
- I know because _____.

Mi afirmación es _____.

Lo sé porque _____.

- So _____.

Entonces _____.

Explain

Light language support

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- **My claim is ____.**
Mi afirmación es ____.
- **My math evidence is ____.**
Mi evidencia matemática es ____.
- **This proves ____ because ____.**
Esto demuestra ____ porque ____.

4. Check Your Explanation

- ☐ I answered the question.
Respondí la pregunta.
- ☐ I used math evidence: numbers, an equation, a model, or a comparison.
Usé evidencia matemática: números, una ecuación, un modelo o una comparación.
- ☐ I explained how the evidence proves my answer.
Explicué cómo la evidencia demuestra mi respuesta.
- ☐ I used at least two math words.
Usé por lo menos dos palabras matemáticas.
- ☐ I reread complete sentences and punctuation.
Volví a leer mis oraciones completas y la puntuación.

Writing Guide — Teacher Copy

The support level changes the amount of language scaffolding, not the mathematical expectation. Look for the same five features in every response.

- I answered the question.
- I used math evidence: numbers, an equation, a model, or a comparison.
- I explained how the evidence proves my answer.
- I used at least two math words.
- I reread complete sentences and punctuation.

Answer Key & Teacher Guide

1. **Notes 1:** Display Data with Histograms
2. **Notes 2:** Histogram
3. **Notes 3:** Frequency
4. **Notes 4:** Interval
5. **Notes 5:** Distribution
6. **Notes 6:** Variability
7. **Watch for this mistake:** *A common mistake in Display Data: Histograms is counting a boundary value in two intervals at once — for example, putting the value 10 in both the '0–9' bin and the '10–19' bin. Each data value belongs to exactly ONE interval. Before you submit, add up all your frequencies and check that the total matches the number of data values you started with.*
8. **Try It 1:** D. 10–19 with 9 players — *The 10–19 interval has the greatest frequency (9 players), so its bar is the tallest.*
9. **Try It 2:** B. 16 — *Add every bar's frequency: $2 + 8 + 5 + 1 = 16$ players total.*